SOUND TO LIGHT

LISTEN · ANALYSE · LIGHT UP

Beginner workshop · 90 minutes

Python + Arduino + addressable LEDs

Processing Community Day · 17 October 2026

Bath Spa University · Locksbrook Road

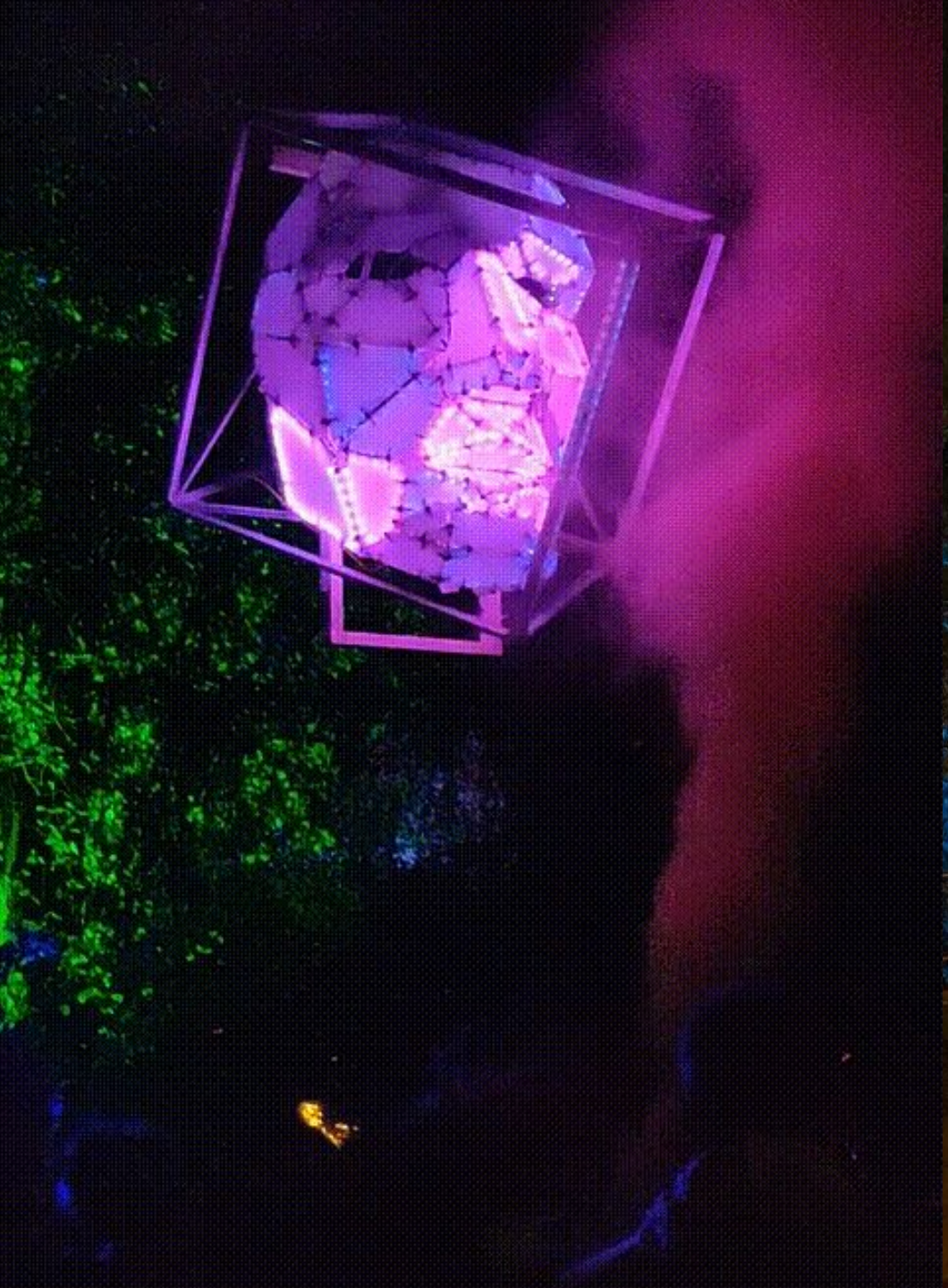
with Ben Winstone - Octopus Immersive



OCTOPUS IMMERSIVE LABORATORY

THE PLAN FOR THE NEXT 90 MINUTES

Time	What we'll do	What you'll have
14:00	Welcome and the big picture	A map of the whole system
14:05	Sound → data → light	Know what data we need
14:15	Meet the toolkit	Everything installed
14:25	FFTs, the friendly version	The idea, no maths
14:35	Hands-on: FFT a file, then your mic	Live numbers on screen
14:50	Send the data over serial	Python talking to USB
15:00	The Arduino reads the data	A blinking onboard LED
15:10	Wire up the LEDs	Lights that dance
15:25	Make it yours, then wrap up	Ideas to take home

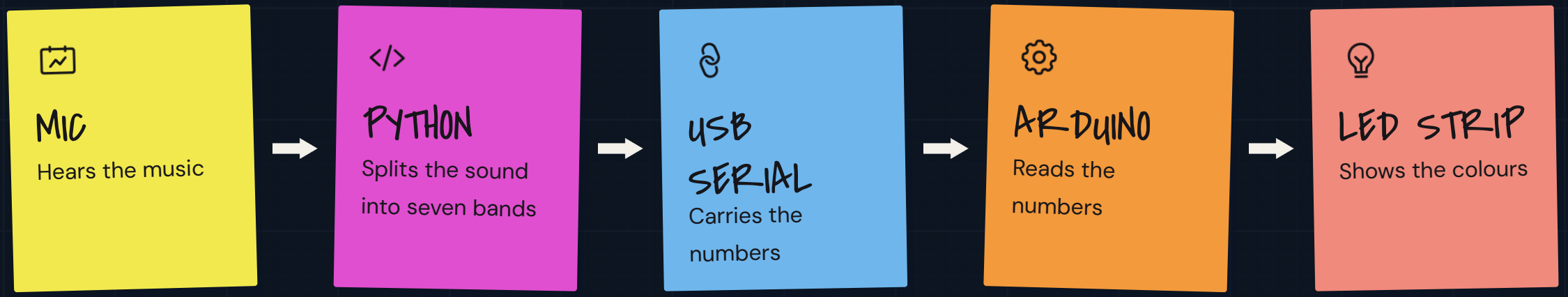




What is it?

The big picture

HERE'S WHAT WE'RE BUILDING TODAY



Your laptop does the thinking. The Arduino does the lighting.

The code is provided for you. Today is more about seeing how things connect together, rather than learning to code - https://github.com/voidnoise/S2L/

What is sound?

SOUND IS AIR, VIBRATING

1

Something vibrates

Over time this creates a waveform or signal

2

The air ripples

It's squeezed and stretched, and that ripple travels to your ear.

3

A mic copies it

It turns the air pressure into a tiny, changing voltage, and these get turned into numbers

Fast wobbles sound high. **Slow** wobbles sound low. **Big** wobbles sound loud.

FOUR TOOLS, EACH WITH ONE JOB

</>

Python

The listener

Listens to the audio and does the maths.

VS Code

THE NOTEBOOK

Where we write and run our Python, and arduino code

Arduino

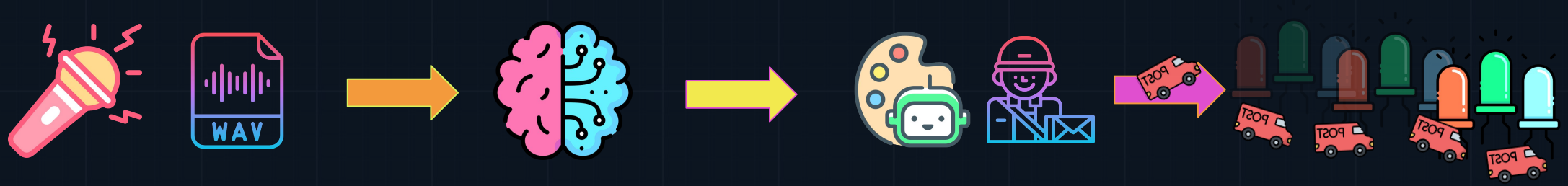
THE MESSENGER

A tiny computer that drives the LEDs.

WS2812 LEDs

THE SHOW

A strip of lights you control one by one.



What does sound data look like?

TO A COMPUTER, SOUND IS JUST NUMBERS



44,100
numbers (samples) every second

-1 to +1
how far the air is pushed or pulled

1,024
samples per chunk we analyse (≈ 23 ms)

How does that translate to lights?

Lights Need Colour Values...

Animations are just LEDs changing colour over time. So changes in sound data can trigger changes in LED colours. But what is the ‘sound data’?

VOLUME

How loud is it overall?

→ **brightness**

BASS

The thump: kick drums, bass lines

→ **pulses**

MIDS

Voices, guitars, melody

→ **colour**

TREBLE

The sparkle: hi-hats, cymbals

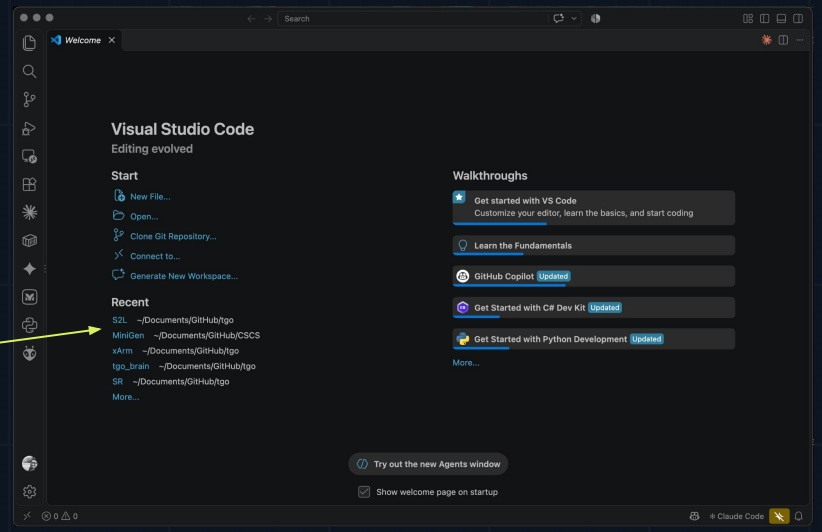
→ **flickers**

Setup check - VS Code

Its like Arduino - but much more versatile

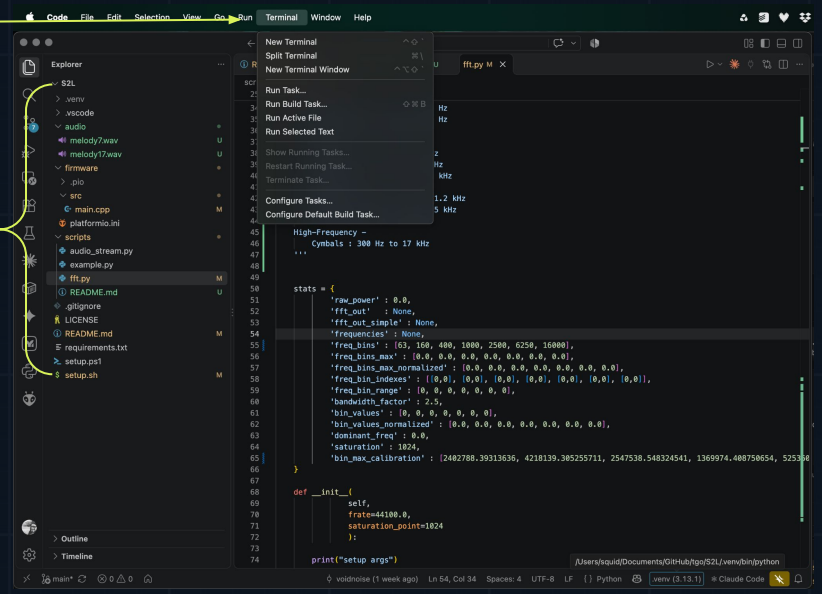
- ✓ **Open VS Code**

  Under Recent -> Open 'S2L'



- ✓ **Open a terminal (command line)**

  Menu bar -> Terminal -> New Terminal



- ✓ **The project folder**

  S2L

  - .venv (python virtual environment to keep things isolated)
  - audio/ (example audio files for testing with)
  - firmware/ (Arduino code)
  - scripts/ (Python code, FFT, serial streaming...)
  - README.md (information / guide)
  - requirements.txt (python dependencies)
  - setup.sh/ps1 (project setup script, should have already been run)

Setup check

BEFORE WE START: IS EVERYTHING INSTALLED?

- ✓ **Python 3**
  In the terminal type python --version in a terminal

- ✓ **Platform IO**
  In the terminal type pio --version in a terminal

Reading The Sound

AN FFT ANSWERS ONE QUESTION: HOW MUCH OF EACH PITCH IS IN THIS SOUND RIGHT NOW?

FAST FOURIER TRANSFORM

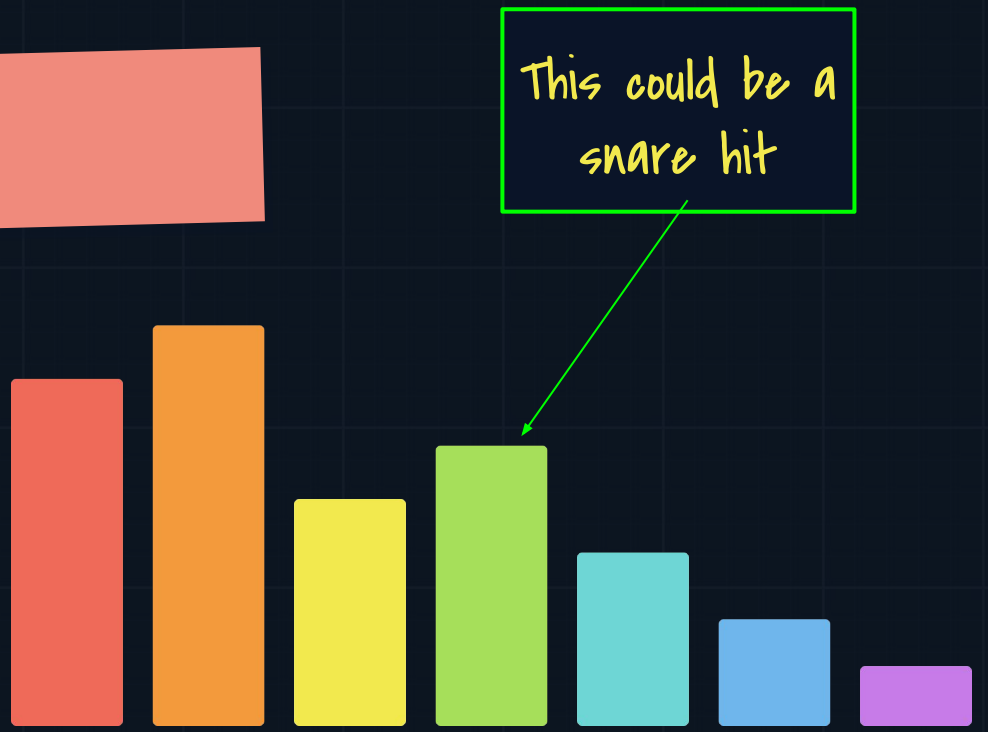
You don't need the maths today — just the idea.

Audio analysis

THINK OF IT AS A SUMMARY OF THE SOUND FOR A MOMENT IN TIME



FFT



In: what the mic hears — one wobbly line, every pitch mixed together

Out: how strong each pitch is, from low to high during a moment in time. It becomes 7 easy numbers, instead of a confusing squiggle.

Audio analysis

WE GROUP THE PITCHES INTO SEVEN BANDS

#	Band	Range	What you hear
0	Sub-bass	20–60 Hz	The rumble you feel
1	Bass	60–250 Hz	Kick drum, bass guitar
2	Low mids	250–500 Hz	Warmth of guitars, low voices
3	Mids	500–2,000 Hz	Most voices and melody
4	Upper mids	2–4 kHz	Snare crack, vocal edge
5	Presence	4–6 kHz	Clarity, “t” and “k” sounds
6	Brilliance	6–20 kHz	Hi-hats, cymbals, “sss”

Hz (hertz) = The frequency/pitch. People hear roughly 20 Hz to 20,000 Hz (20 kHz).

WHAT OUR CODE DOES IN A LOOP

1

Grab a chunk

1,024 samples from the mic or audio file

2

Run the FFT

Python numpy does it in a single line of code.

3

Sum the bands

Squash hundreds of slices into seven bands.

4

Scale and send

Turn each into 0–255 and send it to the Arduino.

Run an FFT on an Audio File

HANDS-ON !

1) In VS Code look at scripts/fft.py
2) In the terminal run:

   python scripts/fft.py -f audio/melody17.wav -d true

   This will play the audio file melody17.wav, whilst grabbing those chunks mentioned previously and passing them through the fft. This then outputs the results to the terminal.
3)
```
~$ python scripts/fft.py -f audio/melody17.wav -d true
playing audio/melody17.wav, ctrl-c to stop
dominant:   893.6 Hz | bins:   22.1   14.9  174.8  260.4  171.5   65.2   48.6
dominant:   893.6 Hz | bins:   11.7    4.9  130.7  177.4  114.6   57.0   34.6
dominant:   893.6 Hz | bins:   33.6   18.7  168.9  206.4  111.5   33.1    9.3
dominant:   699.8 Hz | bins:   77.5   29.8  207.8  236.7  115.2   41.6   27.2
dominant:   699.8 Hz | bins:   77.8   50.4  338.4  353.4  140.6   34.0   26.9
```
4) Instead of passing -f trying -m true, then make sounds to the microphone and see the numbers change

Use a .wav file

Spot the bass

Try the mic input

Addressable LEDs

EVERY LIGHT HAS ITS OWN ADDRESS

Versions of LEDs: NeoPixels, WS2811/12/13/15, SK6812, CS8812/UCS1903/TM1814



ONE DATA WIRE

Colour instructions pass down the chain, LED to LED.

THREE NUMBERS EACH

Red, green and blue, each from 0 to 255.

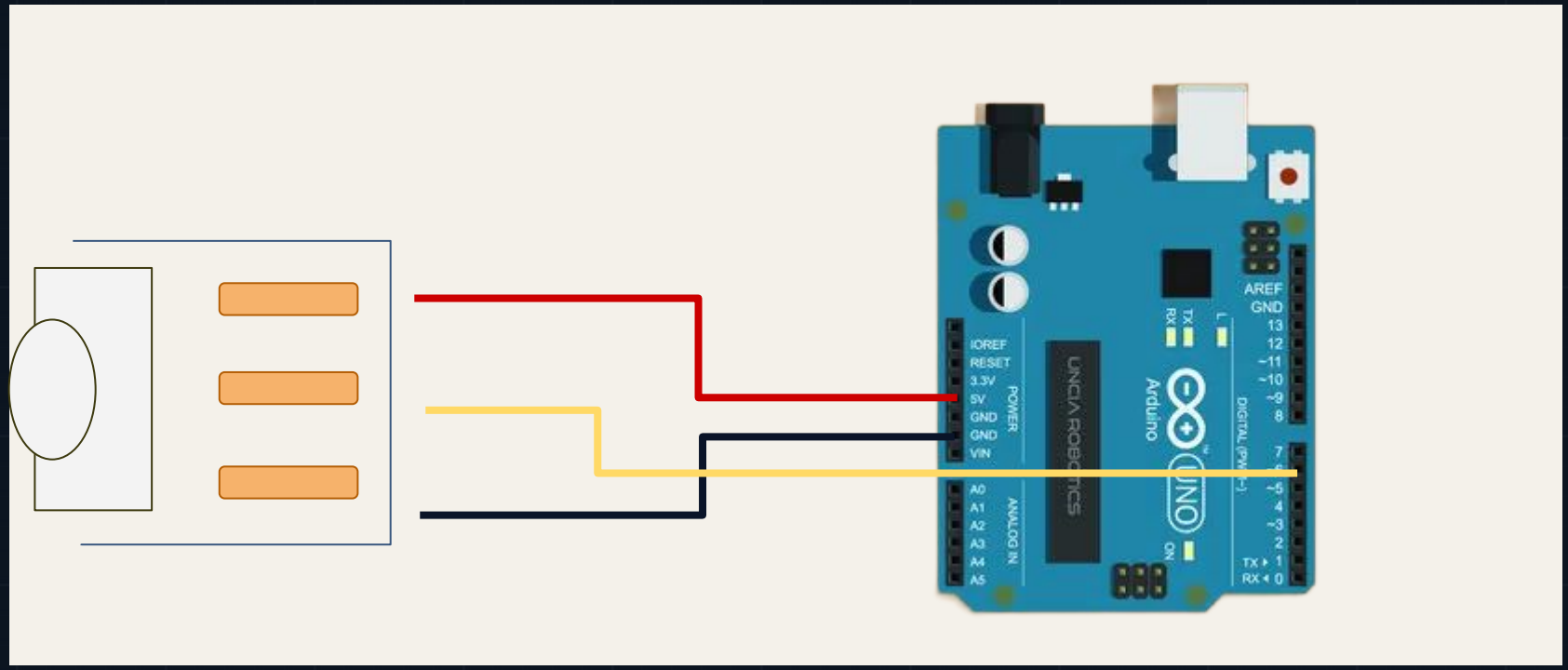
HUNGRY FOR POWER

Up to 60 mA each at full white, so keep brightness low on USB.



HANDS-ON 2

3 wires, 5v (power/Red), Ground (Black), Data (Yellow)



Test the LED wiring

HANDS-ON 2

LET THERE BE LIGHT : led_test

1) In VS Code look for firmware/led_test
2) Download it to the arduino with:

```
pio run -f firmware -e led_test -t upload
```

This will program the arduino to send LED data from pin 6 to help check everything is connected.

Test the communication

HANDS-ON 3

DATA TRANSFER : serial_test & audio_stream.py

1) In VS Code look firmware/serial_test
2) Download it to the arduino with:

```
pio run -f firmware -e led_test -t upload
```

This will program the arduino to send LED data from pin 6 to help check everything is connected.

3) From the scripts folder run:

```
python audio_stream.py --file ../audio/melody7.wav --port
/dev/cu.usbserial-A70064SK --echo
```

This will then output an “echo” message from the Arduino.

HANDS-ON 4

MAKE THE LEDS DANCE : animations

1) In VS Code look firmware/simple_animation
2) Download it to the arduino with:

```
pio run -f firmware -e simple_animation -t upload
```

This will program the arduino to send LED data from pin 6 and drives a simple animation in the LEDs.

3) From the scripts folder run:

```
python audio_stream.py --file ../audio/melody7.wav --port /dev/cu.usbserial-A70064SK
```

This will make the LEDs light up to the music.

HANDS-ON 5

MAKE THE LEDS DANCE : more animations

1) In VS Code look firmware/animations or firmware/animations_fastled
2) Download it to the arduino with:

```
pio run -f firmware -e animations -t upload
   OR
pio run -f firmware -e animations_fastled -t upload
```

This will program the arduino to send LED data from pin 6 and drives more complex in the LEDs.

3) From the scripts folder run:

```
python audio_stream.py --file ../audio/melody7.wav --port /dev/cu.usbserial-A70064SK --animation 2
```

You can send animation 1, 2, or 3 to select which animation is played.

Show Off!

HANDS-ON 6

STRETCH GOAL : MAKE YOUR OWN ANIMATION

1) In VS Code look firmware/animation_fastled/custom.h. This is a basic template to write your own animation. If has some notes, but have an explore and try to link the audio data to the LEDs yourself.
2) Download it to the arduino with:

```
pio run -f firmware -e animations_fastled -t upload
```

3) From the scripts folder run:

```
python audio_stream.py --file ../audio/melody7.wav --port /dev/cu.usbserial-A70064SK --animation 4
```

4) Good luck!

Make it yours

FOUR WAYS TO CHANGE THE ANIMATION

BASS PULSE

Brightness follows the kick drum.

```
FastLED.setBrightness(bands[1] / 4);
```

VU METER

The louder it is, the more LEDs light up.

```
int lit = map(bands[1], 0, 255, 0, NUM_LEDS);
```

COLOUR WHEEL

The mids choose the colour.

```
fill_solid(leds, NUM_LEDS,
  CHSV(bands[3], 255, 255));
```

TREBLE SPARKLES

Random white flashes on the hi-hats.

```
if (bands[6] > 150)
  leds[random8(NUM_LEDS)] = CRGB::White;
```

YOU JUST TURNED SOUND INTO LIGHT !



SOUND → NUMBERS → FFT → SERIAL → ARDUINO → LEDS

Keep going

Thank you!

Code: https://github.com/voidnoise/S2L/

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